11: Inference for MLR

Stat 230 - Spring 2026

```
library(ggformula)
```

Overview

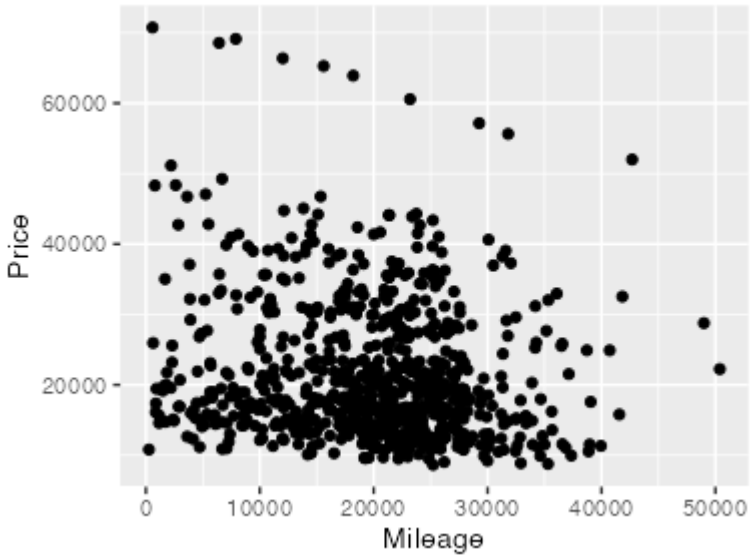
In this activity you will learn how to include build multiple regression models in R, including models with categorical predictors. The `cars` data set loaded below contains information on used cars for sale, including their `Price`, `Mileage`, and `Make` (a categorical variable with six levels).

```
cars <- read.csv("https://aluby.github.io/stat230/data/Cars.csv",
stringsAsFactors = TRUE)
```

Thinking about the data

To begin, create a scatterplot of `Price` versus `Mileage`.

```
gf_point(Price ~ Mileage, data = cars)
```



Q1. What do you notice about the relationship between these two variables? Is a transformation necessary?

There appears to be negative correlation, and some outliers in the Y direction. There may be some curvature. A transformation may help.

Q2. The `Make` variable is categorical with six levels: Buick, Cadillac, Chevrolet, Pontiac, SAAB, and Saturn. To include this variable in a regression model, we need use indicator (i.e., dummy) variables. How many do we need?

There are 6 levels so we need 5 indicator variables

Fitting an MLR model

Build (fit) a multiple regression model using `log(Price)` as the response with `Mileage` and `Make` as the predictor variables. To do this, you use + to separate the predictor variable on the right side of the ~ in the model formula. R will automatically convert a categorical explanatory variable into a set of indicator variables.

```
car_lm <- lm(log(Price) ~ Mileage + Make, data = cars)
summary(car_lm)
```

```
Call:
lm(formula = log(Price) ~ Mileage + Make, data = cars)

Residuals:
     Min       1Q   Median       3Q      Max
-0.53728 -0.14506 -0.02741  0.10750  1.00719

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.009e+01  3.432e-02 293.975  < 2e-16 ***
Mileage       -7.659e-06  1.041e-06  -7.359 4.61e-13 ***
MakeCadillac   6.490e-01  3.814e-02  17.016  < 2e-16 ***
MakeChevrolet -2.956e-01  3.014e-02  -9.809  < 2e-16 ***
MakePontiac   -1.435e-01  3.339e-02  -4.299 1.93e-05 ***
MakeSAAB       3.554e-01  3.516e-02  10.108  < 2e-16 ***
MakeSaturn    -3.972e-01  4.116e-02  -9.650  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.241 on 797 degrees of freedom
Multiple R-squared:  0.6572,    Adjusted R-squared:  0.6546
F-statistic: 254.7 on 6 and 797 DF,  p-value: < 2.2e-16
```

Q3. Report the fitted regression equation and the R^2 value.

$\widehat{\log(\text{Price})} = 10.09 - 7.66 \times 10^{-6} \cdot \text{Mileage} + 0.649 \cdot \text{Cadillac} - 0.296 \cdot \text{Chevrolet} - 0.144 \cdot \text{Pontiac} + 0.355 \cdot \text{SAAB} - 0.397 \cdot \text{Saturn}$

$R^2 = 0.6572$: about 65.7% of the variability in log(Price) is explained by mileage and make.

Q4. Which level of `Make` is the baseline? The baseline level is represented by 0's across all of the indicator variables and won't have a coefficient in the output.

Buick (the category without an indicator variable)

Q5. What strategy did R use to create the indicator variables for `Make`? In other words, how did R decide which level of `Make` to use as the baseline?

Alphabetical order!

Q6. Which of the following models best describes the one just fit: parallel lines, different slopes, or separate lines?

Parallel lines

Plotting the fitted model

To plot a fitted regression model, we can use the `ggpredict()` and `plot()` functions in the {ggeffects} package. The `ggpredict()` function creates a data frame of predicted values from the model for each `Make` across a range of `Mileage` values. The `plot()` function then creates a plot of these predicted values.

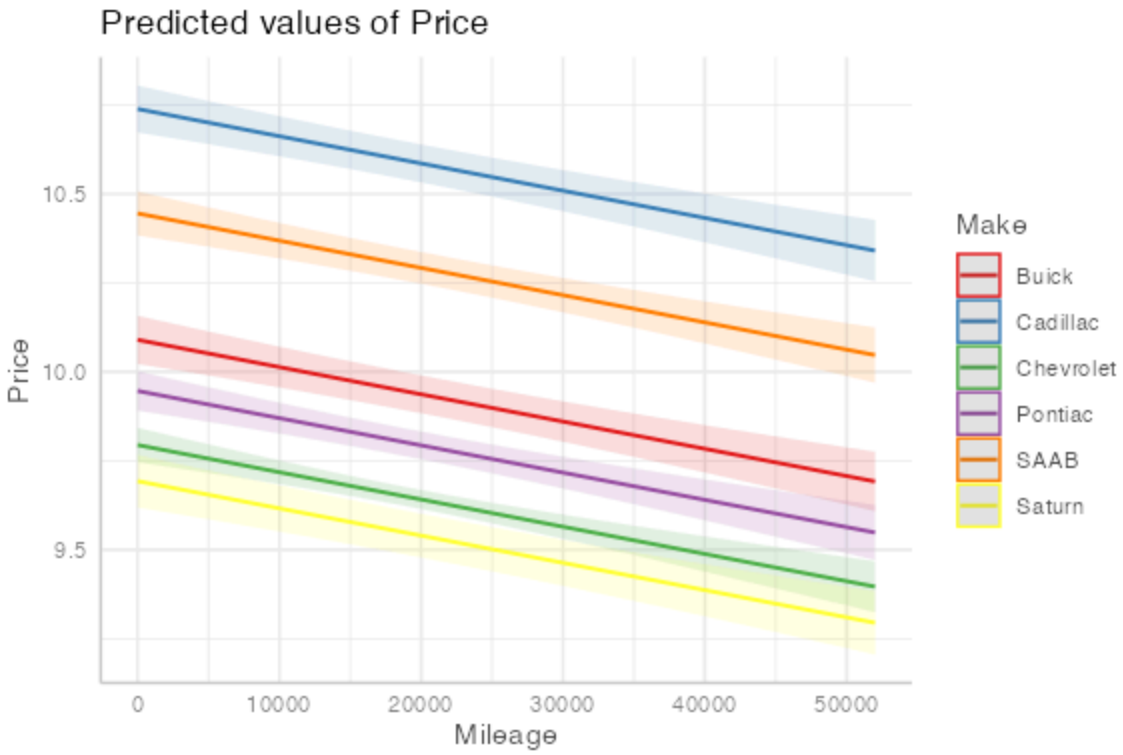
```
library(ggeffects)
cars_pred <- ggpredict(car_lm, terms = ~Mileage + Make)
plot(cars_pred)
```



Q7. Does this plot confirm your answer to the previous question?

It's kind of hard to tell, since the lines are on the original scale. Telling ggpredict to not back-transform makes it clear that yes, this is a parallel lines model.

```
cars_pred <- ggpredict(car_lm, terms = ~Mileage + Make,
                       back_transform = FALSE)
plot(cars_pred)
```



Fitting a model with interactions

If we believe that the association between `Price` and `Mileage` differs by `Make`, then we can include an interaction term in our model. To do this, we use `*` instead of `+` to separate the predictor variables on the right side of the ~ in the model formula. This will include both main effects and the interaction term.

```
car_lm2 <- lm(log(Price) ~ Mileage * Make, data = cars)


summary(car_lm2)
```

```

Call:
lm(formula = log(Price) ~ Mileage * Make, data = cars)

Residuals:
     Min       1Q   Median       3Q      Max
-0.53137 -0.14306 -0.02311  0.10811  0.99917

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)
(Intercept)           1.008e+01  8.447e-02 119.315  < 2e-16 ***
```

```
Mileage               -7.099e-06  3.918e-06  -1.812 0.070409 .
MakeCadillac           6.469e-01  1.056e-01   6.127 1.41e-09 ***
MakeChevrolet         -2.633e-01  9.147e-02  -2.878 0.004106 **
MakePontiac           -1.819e-01  9.871e-02  -1.843 0.065680 .
MakeSAAB               3.561e-01  1.042e-01   3.418 0.000664 ***
MakeSaturn            -3.225e-01  1.174e-01  -2.746 0.006166 **
Mileage:MakeCadillac   1.578e-07  4.953e-06   0.032 0.974593
Mileage:MakeChevrolet -1.623e-06  4.251e-06  -0.382 0.702655
Mileage:MakePontiac    2.019e-06  4.615e-06   0.438 0.661826
Mileage:MakeSAAB      -4.921e-08  4.760e-06  -0.010 0.991754
Mileage:MakeSaturn    -3.673e-06  5.393e-06  -0.681 0.496041
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2414 on 792 degrees of freedom
Multiple R-squared:  0.6582,    Adjusted R-squared:  0.6535
F-statistic: 138.7 on 11 and 792 DF,  p-value: < 2.2e-16
```

Q8. Report the fitted regression equation for Buicks and Cadillacs.

Buick is the baseline, so all of the indicator variables are 0:

$\widehat{\log(\text{Price})} = 10.079 - 7.1 \times 10^{-6} \cdot \text{Mileage}$

For Cadillacs, we add the Cadillac intercept adjustment and the Cadillac slope adjustment:

$\widehat{\log(\text{Price})} = (10.079 + 0.647) + (-7.1 \times 10^{-6} + 1.58 \times 10^{-7}) \cdot \text{Mileage} = 10.725 - 6.94 \times 10^{-6} \cdot \text{Mileage}$

Q9. What is the R^2 value for this model? Does this model appear to fit the data better than the previous model?

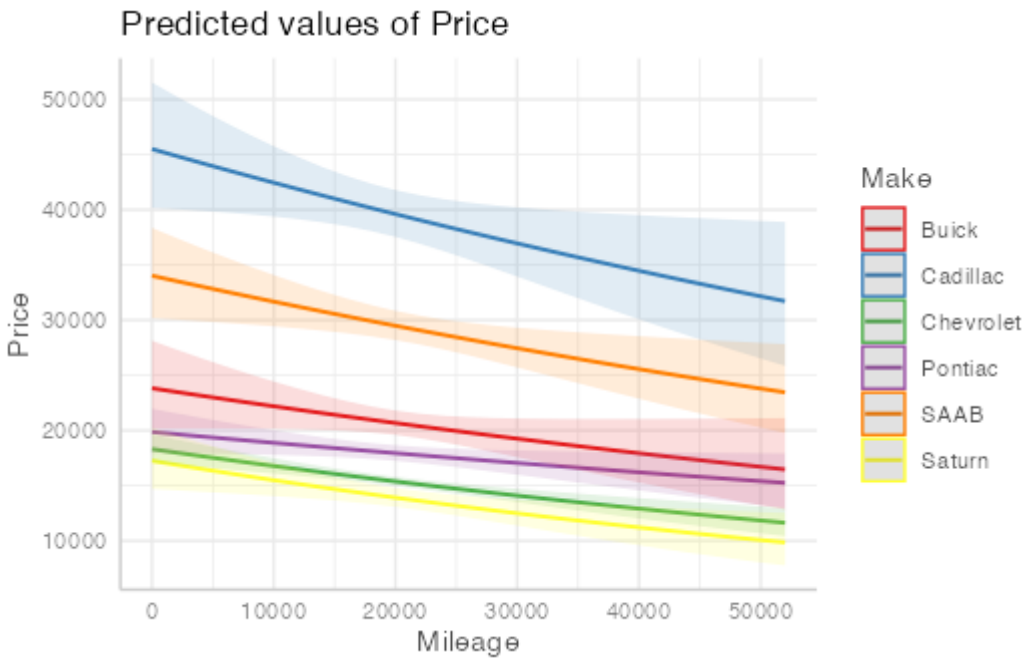
$R^2 = 0.6582$ (compared with 0.6572 for the parallel lines model). R^2 never decreases when we add terms, and it barely changed. The adjusted R^2 actually went *down* (0.6535 vs. 0.6546), so the interaction model does not appear to fit better.

Plotting the fitted model with interactions

To plot the interaction model, we again use the `ggpredict()` and `plot()` functions in the {ggeffects} package.

```
cars_pred2 <- ggpredict(car_lm2, terms = ~Mileage + Make)

# Now plot the predictions
plot(cars_pred2)
```



Calculating extra sums of squares F-tests

Do we really need the interaction terms? To answer this question, we can use an extra sums of squares F-test to compare the model with interaction terms to the model without interaction terms.

We have already seen how to use the `anova()` function to compare two nested models. Here, we will use it again to compare our two car models.

Q10. Fill in the blanks to run the extra sums of squares F-test. What do you conclude about the interaction terms?

```
anova(car_lm, car_lm2)
```

```
Analysis of Variance Table

Model 1: log(Price) ~ Mileage + Make
Model 2: log(Price) ~ Mileage * Make
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    797 46.297
2    792 46.160  5   0.13714 0.4706 0.7983
```

The reduced model (no interaction) goes first. The F-statistic is 0.47 on 5 and 792 degrees of freedom, with a p-value of 0.798. There is no evidence that the interaction terms are needed:

the relationship between `Mileage` and (log) `Price` does not appear to differ by `Make`, so the simpler parallel lines model is sufficient.

Using only the full model

We can also calculate the extra sums of squares F-test using only the full model. To do this, we need to extract the SSE and degrees of freedom for both models from the ANOVA table for the full model.

Q11. Run the below code and verify that the df, sums of squares, mean squares, F value, and p-value match what you got from the previous `anova()` command.

```
anova(car_lm2)
```

```
Analysis of Variance Table

Response: log(Price)
              Df Sum Sq Mean Sq  F value    Pr(>F)
Mileage        1  2.959  2.9591  50.7715 2.336e-12 ***
Make           5 85.802 17.1603 294.4325 < 2.2e-16 ***
Mileage:Make   5  0.137  0.0274   0.4706    0.7983
Residuals    792 46.160  0.0583
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The last row (`Mileage:Make`) matches the extra sums of squares F-test above: the same F-statistic (0.4706), p-value, and degrees of freedom (5). The `Residuals` row gives the SSE and df of the full model, and the sum of squares for the `Mileage:Make` row is the difference between the two models' SSEs (46.297 − 46.16 = 0.137).

Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
`lm()`	Fit a linear model
`summary()`	Display detailed results from a fitted model
`ggpredict()`	Create a data frame of predicted values from a fitted model
`plot()`	Create a plot of predicted values from a fitted model
`anova()`	Create an ANOVA table for a fitted model or compare two nested models